

Delay Problems Questions

Q1] Suppose that the packet length is 4000 bits, and that the link transmission rate along the link from the first router to the second is 10 MBps. What is the transmission delay?

- $L = 4000$ bits
- Transmission Rate, $R = 10 \text{ MBps} = 10 \times 8 \times 10^6 \text{ bps} = 80 \text{ Mbps}$
- Transmission Delay = $L/R = 4000 / (10 \times 8 \times 10^6)$
 $= 0.00005 \text{ s} = 50 \text{ microseconds}$

Q2] There are two ways you can send a 10TB of data from Bengaluru to Chennai. You have a dedicated link for this very purpose transmitting at 100Mbps and a special courier option with delivery in one day. Which would you prefer to use?

- Convert total data to bits
 - $10 \text{ TB} = 10 * 10^{12} * 8 \text{ bits} = 8 \times 10^{13} \text{ bits}$
- Using the dedicated link, transmission time will take
$$10 * 10^{12} * 8 / (100 * 10^6) = 800,000 \text{ seconds} / 86,400 \sim 9.26 \text{ days.}$$

Network = 9.3 days

But with the special courier option, it would arrive in a single day. Hence the courier option would be preferred for speedy delivery.

Q3] An optical fiber cable is able to transfer packets across the network at a speed of 70% of the speed of light in 10 microseconds. What is the distance the packet traverses?

- Distance = Speed x Time
- Speed of propagation = 70% of 3×10^8 m/s
= $0.7 \times 3 \times 10^8$
= 2.1×10^8 m/s
- Propagation delay = d/s
- Distance traversed = Propagation delay x speed of propagation
= $2.1 \times 10^8 \times 10 \times 10^{-6} = 2100$ m = 2.1km

Q4] The time taken by a packet to travel across a link is 1000 times the time taken for the packet to be inserted in the link. Assuming no queuing and processing delay, what is the transmission delay of the link if it connects an end to end distance of 2km in the network in a total of 3 microseconds ?

- Given: Distance = 2 km, Total delay = 3 μ s
- Propagation delay = 1000 $\times$ transmission delay
- Assuming no queuing and processing delay,
- End to end delay = Transmission delay + Propagation delay
- End-to-end delay:
- 3 micro = Trans delay + Propagation delay = TD + (1000 TD) = 1001TD
- Solving gives:
- Transmission delay $\sim$ 2.997 nano sec $\sim$ 3ns
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Q5] Data traveling in Alice's network of 2 computers, takes a one hop transmission delay of 0.003ms to go from one computer to the other. If the queuing delay is 1 microsecond and the distance between the two systems is 3km, find the time taken by Alice's router for bit level error checking and processing if the total nodal delay is 0.04ms.

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

$$0.04 \times 10^{-3} = d_{\text{proc}} + 0.001 \times 10^{-3} + 0.003 \times 10^{-3} + 3000 / (3 \times 10^8)$$

Processing delay,

$$d_{\text{proc}} = (0.04 - 0.01 - 0.001 - 0.003) \times 10^{-3} = 0.026 \text{ ms}$$

Throughput Problems

Q6] Hosts Alice and Bob are connected to each other via a router. The bandwidth of each link between Alice and the router is 15Mbps. If the efficiency of data transfer in the entire network is 80%, what is the actual throughput of the network?

- Ideal Throughput = $\min\{15, 15\} = 15\text{M bps}$
- Efficiency = Amount of data transferred through the network per second / Amount of data that could be transferred per second
 $80 / 100 * 15$
- Throughput(x) = 12M bps

Q7] Consider a 10 Mbps link being shared by 7 users. What is the maximum throughput if the network is using circuit switching?

- In **circuit switching**, the link bandwidth is **divided equally** among all users.
- **Maximum throughput per user:**
- Throughput *per user* = $\frac{10 \text{ Mbps}}{7}$
- = 1.4286 Mbps
- $\approx 1.43 \text{ Mbps}$
- The throughput is nearly equal to half of the bandwidth.

Q8] Alice and Bob are connected to each other via a router. The bandwidth of each link between the router and Bob is 20Mbps. If the network allows for the actual transmission of data at 20Mbps, then what is the time taken for a packet of size 1000 bits to enter the network?

- Transmission Delay=Packet Size /Transmission Rate = L/R
- Packet Size = 1000, $R = 20\text{Mbps}$
- Delay= 20×10^6
- $= \frac{1000}{20,000,000}$
- $= 5 \times 10^{-5}$ seconds
- = 50 microsec

Q9] This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

- a. Express the propagation delay, d_{prop} , in terms of m and s.

Propagation delay is:

$$d_{prop} = \frac{\text{Distance}}{\text{Propagation speed}}$$
$$d_{prop} = \frac{m}{s}$$

- b. Determine the transmission time of the packet, d_{trans} , in terms of L and R.

Transmission delay is the time required to push all L bits onto the link:

$$d_{trans} = \frac{\text{Packet size}}{\text{Transmission rate}}$$
$$d_{trans} = \frac{L}{R}$$

c. Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.

Total delay = $d_{trans} + d_{prop}$

$$d_{end-end} = \frac{L}{R} + \frac{m}{s} \text{ sec}$$

d. Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{trans}$, where is the last bit of the packet?

At $t = d_{trans}$, the sender has just finished transmitting the last bit.

So:

The last bit is just leaving Host A (at the beginning of the link).

e. Suppose d_{prop} is greater than d_{trans} . At time $t = d_{trans}$, where is the first bit of the packet?

The first bit is in the link and has not reached Host B.

f. Suppose d_{prop} is less than d_{trans} . At time $t = d_{trans}$, where is the first bit of the packet?

$d_{prop} < d_{trans}$.

- If propagation delay is smaller:
- The first bit reached Host B at time d_{prop} , which is earlier.

The first bit has already arrived at Host B.

g. Suppose $s = 2.5 \times 10^8$, $L = 1500$ bytes, and $R = 10$ Mbps. Find the distance m so that d_{prop} equals d_{trans} .

$$s = 2.5 \times 10^8 \text{ m/s}$$

- $L = 1500 \text{ bytes} = 1500 \times 8 = 12000 \text{ bits}$
- $R = 10 \text{ Mbps} = 10 \times 10^6 \text{ bps}$
- We are asked to find distance m such that: $d_{\text{prop}} = d_{\text{trans}}$
- **Compute transmission delay**
- $d_{\text{trans}} = \frac{12000}{10 \times 10^6} = 1.2 \times 10^{-3} \text{ seconds} = 0.0012 \text{ s}$
- **Set equal to propagation delay**
- $\frac{m}{s} = 0.0012$
- $m = s \cdot 0.0012$
- $m = (2.5 \times 10^8)(0.0012)$
- $m = 3 \times 10^5 \text{ meters}$
- $m = 300,000 \text{ meters} = 300 \text{ km}$

Q10] Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected by a direct link of $R = 5$ Mbps. Suppose the propagation speed over the link is 2.5×10^8 meters/sec.

- Calculate the bandwidth-delay product, $R \cdot d_{\text{prop}}$.
- Distance = 20,000 km = 2×10^7 m
- $R = 5$ Mbps = 5×10^6 bps; Propagation speed $s = 2.5 \times 10^8$ m/s
- $d_{\text{prop}} = \frac{2 \times 10^7}{2.5 \times 10^8} = 0.08$ seconds
- $R \cdot d_{\text{prop}} = (5 \times 10^6)(0.08) = 4 \times 10^5$ bits = 400,000 bits

- Consider sending a file of 800,000 bits from Host A to Host B. Suppose the file is sent continuously as one large message. What is the maximum number of bits that will be in the link at any given time?

If the file is sent continuously, the maximum number of bits that can be “in flight” equals the bandwidth-delay product.

Since file size = 800,000 bits (which is larger than 400,000),

$$\text{Maximum } \textit{bits in link} = 400,000 \textit{ bits}$$

- Provide an interpretation of the bandwidth-delay product.

The bandwidth-delay product represents:

The maximum number of bits that can be in transit (in the link) at any given time.

- What is the width (in meters) of a bit in the link? Is it longer than a football field?

First find transmission time of one bit:

$$\text{Time per bit} = \frac{1}{R} = \frac{1}{5 \times 10^6} = 2 \times 10^{-7} \text{ sec}$$

Now multiply by propagation speed:

$$\text{Bit width} = s \times \frac{1}{R} = (2.5 \times 10^8)(2 \times 10^{-7}) = 50 \text{ meters}$$

A football field $\approx$ 100 meters.

$$50 \text{ m} < 100 \text{ m}$$

No, it is about half a football field.

Q10] For the above problem, now with a link of $R = 1$ Gbps.

a. Calculate the bandwidth-delay product, $R \cdot d_{\text{prop}}$.

$$d_{\text{prop}} = m/s = 2 \times 10^7 / 2.5 \times 10^8 = 0.08 \text{ s}$$

$$R \cdot d_{\text{prop}} = (10^9)(0.08) = 8 \times 10^7 = 80,000,000 \text{ bits}$$

80 million bits (80 Mb)

b. Consider sending a file of 800,000 bits from Host A to Host B. Suppose the file is sent continuously as one big message. What is the maximum number of bits that will be in the link at any given time?

File size = **800,000 bits**

Bandwidth-delay product = **80,000,000 bits**

Since the file size is much smaller than the pipe size:

Maximum bits in the link = 800,000 bits

c. What is the width (in meters) of a bit in the link?

Width of one bit:

$$\begin{aligned} \text{Bit width} &= \frac{s}{R} \\ &= \frac{2.5 \times 10^8}{10^9} \\ &= 0.25 \text{ meters} \end{aligned}$$

0.25 meters (25 cm)

Q11] An ISP offers internet connections with transmission speeds of 10 Mbps. To attract customers, it promises less than 5 ms of end-to-end delay while playing online games, if the other players are also their customers and are no more than 10,000 km away. Assume an idle network with propagation speed of 2.5×10^8 m/s and no processing and queuing delays. If the average amount of data generated per second is 100 Kilo Bytes, find out if the ISP can deliver on their promise.

- Transmission rate $R = 10 \text{ Mbps} = 10 \times 10^6 \text{ bps}$
- Distance $m = 10,000 \text{ m} = 10,000 \text{ km} = 10^7 \text{ m}$
- Propagation speed $s = 2.5 \times 10^8 \text{ m/s}$
- No processing or queuing delay
- Data generated per second = **100 KB/s**
- **Propagation Delay**
 - $d_{\text{prop}} = m/s = 10^7 / 2.5 \times 10^8 = 0.04 \text{ seconds} = 40 \text{ ms}$
- **Transmission Delay**
 - 100 KB per second: $= 100 \times 1000 = 100,000 \text{ bytes/sec}$
 - Convert to bits: $= 100,000 \times 8 = 800,000 \text{ bits/sec}$
 - Transmission delay per second of data: $d_{\text{trans}} = \frac{800,000}{10 \times 10^6} = 0.08 \text{ seconds} = 80 \text{ ms}$
 - $d_{\text{total}} = d_{\text{prop}} + d_{\text{trans}} = 40 + 80 = 120 \text{ ms}$

Promised delay: **< 5 ms**

Actual delay: **≈ 120 ms.**

The ISP cannot deliver on their promise.

Q12] Two hosts are separated by a distance of 5,000 km. Two packets that are 4 Mega Bytes in size must be sent from one host to the other. The queuing delay at the host is 10 ms and the propagation speed of the link is 2.5×10^8 m/s. Find the transmission speed at the source such that it finishes transmitting the second packet when the first packet just arrives at the destination.

Distance $m = 5000 \text{ km} = 5 \times 10^6 \text{ m}$

Packet size = **4 MB**

Number of packets = 2

Queuing delay = 10 ms

Propagation speed $s = 2.5 \times 10^8 \text{ m/s}$

Propagation Delay = $d_{\text{prop}} = d/s = 5 \times 10^6 / 2.5 \times 10^8 = 0.02 \text{ sec} = 20 \text{ ms}$

Convert Packet Size = $4 \text{ MB} = 4 \times 10^6 \text{ bytes} = 4 \times 10^6 \times 8 = 32 \times 10^6 \text{ bits}$

The source finishes transmitting the second packet exactly when the first packet arrives.

When does the first packet arrive?

The first packet must:

1. Wait 10 ms in queue
2. Be transmitted $\rightarrow d_{\text{trans}}$
3. Propagate $\rightarrow 20 \text{ ms}$

Time when first packet arrives: $d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}} = 10 + d_{\text{trans}} + 20$
 $= 30 + d_{\text{trans}} \text{ (in ms)}$

- **When does the second packet finish transmitting?**

The second packet:

Waits 10 ms in queue

First packet is transmitted $\rightarrow d_{trans}$

Then second packet is transmitted $\rightarrow$ another d_{trans}

Time when second packet finishes transmission:

$$\begin{aligned}d_{queue} + 2d_{trans} \\ = 10 + 2d_{trans}\end{aligned}$$

For the second packet to finish transmitting exactly when the first arrives:

Transmission delay must equal propagation delay.

Finish transmitting second packet = Arrival of first packet

$$10 + 2d_{trans} = 30 + d_{trans}$$

$$\begin{aligned}2d_{trans} - d_{trans} &= 30 - 10 \\ d_{trans} &= 20 \text{ ms}\end{aligned}$$

Transmission Rate:

$$d_{trans} = L/R$$

$$0.02 = \frac{32 \times 10^6}{R}$$

$$R = \frac{32 \times 10^6}{0.02}$$

$$R = 1.6 \times 10^9 \text{ bps}$$

$$R = 1.6 \text{ Gbps}$$

Q13] 100 Tera Bytes of data is to be transferred between two hosts, Host A in Bengaluru and Host B in Chennai, separated by a distance of 350 km. The transmission speed at Host A is 1 Gbps and the propagation speed of the dedicated link is 2×10^8 m/s. Which one of these is better:

- to transfer the data over the link or
- to drive to Chennai to deliver the data and drive back to Bengaluru, if a one-way trip takes 7 hours

Which component is dominant in the total time taken for the data transmission?
Assume negligible processing and queuing delays.

Data size = **100 TB**, Distance = **350 km**

- Transmission rate $R = 1 \text{ Gbps} = 10^9 \text{ bps}$
- Propagation speed $s = 2 \times 10^8 \text{ m/s}$
- One-way drive time = **7 hours**
- Negligible processing and queuing delays
- Convert Data to bits: $= 100 \times 10^{12} \times 8 = 8 \times 10^{14} \text{ bits}$
- **Transmission Time:** $d_{trans} = \frac{8 \times 10^{14}}{10^9} = 8 \times 10^5 \text{ second} = 800,000 \text{ seconds}$
- Convert to hours: $\frac{800,000}{3600} \approx 222 \text{ hours} \approx 9.25 \text{ days}$

- **Propagation Delay**

$$d_{prop} = \frac{350 \times 10^3}{2 \times 10^8} = 0.00175 \text{ sec} = 1.75 \text{ ms}$$

This is negligible compared to days.

Total Network time = 9.25 days

Round Trip time (driving) = 7 + 7 = 14hrs

Driving is much faster

- **Dominant Component:** The **Transmission Delay** (d_{trans}) is the dominant component. While the propagation delay is a mere 1.75 msec, the time required to push 100 TB of data through a 1 Gbps "pipe" is so massive that it accounts for 99.99% of the total time.

Q14] There are two paths to transmit 100 packets each of size 16 Mega Bytes from host A to host B: a lossy channel with loss probability of 50% and transmission speed of 20Gbps; and a lossless channel with transmission speed of 10 Gbps. If both the channels have a propagation speed of 2×10^8 m/s and the queuing delay at the host is 10 ms per packet, which path is better? Assume all lost packets are re-transmitted with no additional overhead.

- **Convert Packet Size = 16MB = $16 \times 8 \times 10^6 = 128$ Mbits**

- **Lossless (10 Gbps)**

- **Transmission delay per packet**

- $d_{trans} = \frac{128 \times 10^6}{10 \times 10^9} = 0.0128 \text{ s} = 12.8 \text{ ms}$

- **Total delay per packet**

- $10 + 12.8 = 22.8 \text{ ms}$

- **For 100 packets**

- $100 \times 22.8 = 2280 \text{ ms} = \boxed{2.28 \text{ seconds}}$

- **Path 2: Lossy (20 Gbps, 50% loss)**

Expected transmissions per packet:

$$\frac{1}{1 - 0.5} = 2$$

So total transmissions:

$$100 \times 2 = 200$$

- **Transmission delay per packet**

$$\frac{128 \times 10^6}{20 \times 10^9} = 0.0064 \text{ s} = 6.4 \text{ ms}$$

- **Total per transmission:** $10 + 6.4 = 16.4 \text{ ms}$

- **For 200 transmissions**

$$200 \times 16.4 = 3280 \text{ ms} = \boxed{3.28 \text{ seconds}}$$

Even though the lossy link has double bandwidth, the 50% retransmissions double the number of transmissions and increase queuing delay significantly.

Use the lossless 10 Gbps channel.